

JOINT PRECISION BUDGETING ACROSS ATTENTION KV AND RECURRENT STATE IN HYBRID LINEAR-ATTENTION SERVING

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ABSTRACT

Hybrid linear-attention models pair a small set of GQA attention layers with interleaved Gated Delta Network (GDN) layers, so serving memory is split between a growing, quantizable attention KV cache and a fixed per-sequence recurrent state that no KV quantizer touches. Existing systems optimize the KV bit-width and leave state precision fixed. We treat state precision as a second budget dimension and implement a selector that maps quality, memory, concurrency, and latency constraints to executable precision configurations. On Qwen3.5-2B/9B served with vLLM on one RTX 5090, a reproducible 112-cell allocator matrix shows higher capacity in all 52 paired fp32/bf16 cells, with a median gain of 15.44%. The idealized capacity model has 1.81% median absolute residual (range -3.66% to $+13.21\%$); it is a prediction, not a bound. Quality remains task-dependent: GSM8K loses 1.0 percentage point at 2B, whereas five preregistered no-think RULER cells show exact observed agreement across three dataset seeds, with all 30 accuracy values and sample outputs reproduced exactly. This equality does not establish equivalence. The selector reproducibly maps three preregistered budgets to full, state-only, and joint precision, but only for one 2B/4K Random-workload slice. Controlled mechanism contrasts reproduce throughput, not tail latency. A broader four-configuration serving matrix completes 144/144 cells per attempt but fails its continuous-goodput stability gate (183/720 comparisons outside 10% tolerance). We therefore claim a verified capacity effect and scoped policy execution, while reporting serving instability rather than a universal performance benefit.

1 INTRODUCTION

KV caches are the dominant memory consumer in LLM serving, and KV quantization is the standard lever for fitting more context and concurrency into a fixed GPU budget (Liu et al., 2024; Hooper et al., 2024; Sharma et al., 2024). The field has largely been built on attention-only transformers, in which every layer contributes a growing, homogeneous, quantizable cache.

Hybrid linear-attention models break that assumption. Qwen3.5, the subject of this paper, interleaves 18 (2B) or 24 (9B) Gated Delta Network (GDN) layers (Yang et al., 2025) with 6 (2B) or 8 (9B) GQA attention layers. The GDN layers keep no per-token KV; instead each sequence carries a fixed recurrent state that is allocated from the same GPU memory pool as the attention KV. Consequently the serving memory footprint of a hybrid model is two-dimensional: a per-token component that KV quantization can shrink, and a per-sequence state component whose size depends on

its own data type. State precision is a deployment switch in vLLM (`mamba_ssm.cache_dtype`), but existing budget analyses, including prior work on state compression itself (Dao, 2026; vLLM Project, 2026), do not treat it as a dimension that should be jointly allocated with the KV bit-width.

This paper is a system study with four contributions. (a) A capacity model that separates concurrent sequences $N(L) = M/(AL + G)$ from total token capacity $T(L) = LN(L)$, plus a deterministic probe validated by two complete 112-cell attempts. (b) An executable joint-precision selector that filters full, KV-only, state-only, and joint candidates by quality, memory, concurrency, and tail-latency constraints, then maximizes a robust SLO-goodput lower bound and fails closed on unmeasured strata. (c) A paired quality map across perplexity, GSM8K, and no-think RULER, with statistical resolution and reproduction status stated per endpoint. (d) Controlled mechanism contrasts and a four-configuration serving matrix whose partial and failed reproducibility results are retained rather than converted into a performance headline. The target regime is short-context, high-concurrency, memory-bound serving, where the fixed state is a large fraction of the budget.

The headline result is a verified capacity effect. Across

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the frozen matrix of model size, KV dtype, context length, and memory utilization, all 52 fp32/bf16 pairs favor bf16 state and the median gain is 15.44%. The clean R2-to-R3 comparison reproduces all 112 cells within the preregistered tolerances. The architecture-derived model has a median absolute residual of 1.81%, with signed residuals from -3.66% to $+13.21\%$; page and block rounding therefore make it an idealized predictor rather than a lower or upper bound. In a representative 2B/4K cell, the int4-over-fp16 KV capacity ratio rises from 2.245 to 2.675 when state is also bf16.

The quality evidence is bounded. Perplexity changes are within $[-0.013, +0.007]$ (C4) and $[-0.045, +0.058]$ (PG19) with all confidence intervals containing zero. GSM8K (200 seeded items, 9 seeds) shows a significant 1.0-point regression at 2B state-bf16 ($p = 0.025$), which we report together with its pre-registered minimum detectable effect of 1.16 points and 67.5% observed power; the 9B model shows no regression. In the no-think RULER repair, every tested fp32/bf16 pair has a delta of 0.00 across three dataset seeds, and an independent temporal attempt reproduces all 30 accuracy values and sample outputs exactly. These are observed agreements, not equivalence or non-inferiority results. The controller’s three mappings also reproduce within their frozen tolerances. In contrast, tail-latency mechanism metrics and the broader four-configuration goodput matrix do not pass their respective stability gates, so we make no general serving-speed claim.

2 BACKGROUND AND RELATED WORK

KV cache quantization and eviction. KIVI (Liu et al., 2024) and KVQuant (Hooper et al., 2024) established 2–4-bit per-token-head KV quantization; MiniKV (Sharma et al., 2024) co-designs 2-bit layer-discriminative KV quantization with the vLLM system, while TurboQuant (Zandieh et al., 2026) applies online vector quantization to the KV cache at 2–4 bits per entry in vLLM. Neither work treats the recurrent-state dtype as an allocatable dimension. Eviction and hybrid policies (H2O (Zhang et al., 2023), StreamingLLM (Xiao et al., 2024), SnapKV (Li et al., 2024)) release memory by dropping tokens rather than reducing precision. All of these target attention-only models. On a hybrid model they compress only the attention layers; the GDN state remains untouched. Eviction and offloading therefore operate on a different axis from the state-dtype switch: they change which tokens or layers stay in memory, whereas state-bf16 changes the bytes per recurrent-state element. Our formal matrix holds eviction and offloading fixed, so the two levers are not mixed in the measured deltas. Recent system-level work also co-optimizes eviction with quantization within the attention-KV budget (HqKV (Wang et al., 2026), RDKV (Zhang et al., 2026), ARKV (Lei &

Ilager, 2026)).

Hybrid linear-attention architectures. Mamba (Gu & Dao, 2024) and Mamba2 (Dao & Gu, 2024) established selective state-space layers; Gated Delta Networks (Yang et al., 2025) make the recurrence a delta rule. Production hybrids (Jamba (Lieber et al., 2024), RecurrentGemma (Botev et al., 2024), Zamba (Glorioso et al., 2024), Qwen3.5 (Team, 2026)) interleave such layers with a small number of full-attention layers. RecurrentGemma builds on Griffin (De et al., 2024); the GDN recurrence continues the delta-rule lineage of DeltaNet (Yang et al., 2024). Serving them requires a per-sequence state in addition to the attention KV; vLLM’s PagedAttention memory manager (Kwon et al., 2023) now allocates both from the same GPU pool.

State precision and compression. Quantizing SSM states has been explored in the low-precision inference literature: ReplaySSM proposes caching inputs instead of state, and vLLM exposes a configurable state-dtype path (Dao, 2026; vLLM Project, 2025); FP8 state checkpointing is under development in the FlashInfer SSU path (vLLM Project, 2026). Recurrent state caching also appears in inference frameworks such as FlashInfer (Ye et al., 2025). Weight-and-activation quantization of SSM layers (Quamba (Chiang et al., 2024), MambaQuant (Xu et al., 2025)) targets the compute path with W8A8 or W4A8 schemes and does not vary the per-sequence state dtype that serving allocators budget, which is the dimension we vary. QPruningKV (Zhang et al., 2025) characterized the token-precision trade-off under a fixed byte budget for attention models. State compression itself is prior art. The gap is joint budget accounting: serving systems expose state dtype without analyzing it as a precision dimension that multiplies with the KV bit-width decision, and without validating the resulting capacity-quality-serving trade-off with disclosed statistical resolution and a second formal run.

3 PRECISION BUDGET MODEL

Let L be the context length, A_f and A_q the per-token attention-KV bytes at fp16 and int4 KV respectively, and G the per-sequence state bytes at the chosen state dtype. With M bytes available for KV plus state, the maximum number of concurrent length- L sequences and the corresponding total token capacity are

$$N(L) = \frac{M}{A \cdot L + G}, \quad T(L) = LN(L) = \frac{L \cdot M}{A \cdot L + G}, \quad (1)$$

where $A \in \{A_f, A_q\}$ and G depends on the state dtype; the denominator is the per-sequence footprint of a length- L request. The capacity ratio from switching the state fp32 to bf16 is

$$r_{state}(L) = \frac{AL + G_{fp32}}{AL + G_{bf16}}, \quad (2)$$

which decreases with L : state dominates at short context, KV dominates at long context. The corresponding KV ratio at a fixed state dtype is $r_{kv}(L) = (A_f L + G)/(A_q L + G)$. The parameters A_q , A_f , and G are derived from the model configuration and the vLLM page layout (per-layer state shapes and padded block sizes), independent of the probed token counts, so the comparison of Eq. (2) against measurements is not circular. Details of the derivation and the padded block accounting are in the appendix.

For deployment stratum x and candidate precision configuration q , the selector implements

$$q^*(x) = \arg \max_{q \in \mathcal{F}(x)} \text{Goodput}(q, x), \quad (3)$$

where $\mathcal{F}(x)$ contains only measured candidates satisfying $\underline{\Delta Q}_t \geq -\epsilon_t$ for every requested task, $\overline{\text{TFTT}}_{95} \leq \tau_f$, $\overline{\text{TPOT}}_{95} \leq \tau_p$, cache bytes $\leq M$, and $N_q(L) \geq N_{min}$. Underlines and overlines denote the stored lower and upper confidence bounds. Missing or ambiguous profile rows yield no feasible candidate rather than interpolation or extrapolation.

4 EVALUATION SETUP

Platform and engine. All experiments use Qwen3.5-2B or Qwen3.5-9B on a single RTX 5090 (32 GB). Earlier serving and quality matrices use a frozen vLLM worktree at commit 55f4768; the preregistered capacity, repaired RULER, controller, mechanism, and four-configuration packages use the later frozen worktree at commit e2fa285. Memory utilization is a protocol-level parameter rather than a global constant: 0.85 for the earlier matrices and RULER, 0.70–0.90 in the capacity grid (plus 0.85 controls), and 0.80 for the controller and follow-up serving packages. Each package archives its resolved cache dtypes and protocol fields, and repetitions are compared only within the corresponding frozen package.

Capacity probes. We use the engine’s maximum-token probe with 1K–32K contexts, uniform int4 KV (int4_per_token_head) or fp16 KV, state fp32/bf16, and the preregistered utilization grid. Each cell is a single deterministic allocator run; each attempt records the resolved dtype, block counts, and serving configuration. Two clean attempts each complete 112/112 cells (52 fp32/bf16 core pairs plus 8 fp16-state controls), and Gate 4 compares per-cell tokens, directions, and model-by-KV median gains. Table 1 shows an independently verified fixed-utilization 4K/16K slice used for deployment interpretation.

Quality. Perplexity is measured with a chunk-level harness (write-back rounding at chunk boundaries, chunk 128) on C4 and PG19, 5 sequences, 3 dataset seeds; we report in Section 6 why this harness is an approximation. Reasoning quality uses the vLLM kernel path: GSM8K (Cobbe

et al., 2021) with 200 seeded items per seed, greedy, no chain-of-thought, 9 seeds; RULER (Hsieh et al., 2024) on five preregistered model-task-length cells, 20 samples per cell, 3 dataset seeds, fixed engine seed, thinking disabled, and a 256-token generation cap. All comparisons are paired across dataset seeds. For RULER, all paired deltas are identical zeros, so the zero-width descriptive intervals denote observed equality only.

Joint-precision controller. Each deployment request specifies the model and context, workload and offered load, memory and concurrency budgets, TTFT/TPOT SLOs, and a task-level quality floor. The controller requires an exact measured profile row for every candidate and does not interpolate across strata. It rejects candidates whose quality lower bound, cache bytes, concurrency, or latency upper bounds violate the request, maximizes the remaining SLO-goodput lower confidence bound, and maps the selected KV/state dtypes to a cold-restart vLLM command. We test three frozen budgets on Qwen3.5-2B at 4K, Random workload, TP=1, with seeds 11, 23, and 47. The parent and temporal reproduction each contain 9 samples and 18,000 measurement requests, with failures retained in the SLO denominator.

Serving. Protocol v3: piecewise constant Poisson arrivals, 120 s warmup, 60 s (300 s for boundary confirmation) measurement windows, TTFT thresholds 250–3000 ms, TPOT 200 ms, goodput/offered ≥ 0.95 required for a sustainable boundary, failures counted in the denominator, 3 seeds. Workloads: Random60 (60-token requests) and ShareGPT300 (300-token traces (LMSYS, 2023)). The formal matrix was followed by a second formal run with the same contracts and seeds (run-stability evidence, not an independent sample); both are committed as atomic artifacts (MLSys, 2026).

Mechanism and four-configuration contracts. Mechanism isolation crosses full and joint precision with fixed block count, fixed cache bytes, and fixed concurrency (3 seeds; 18 cells and 5,400 requests per attempt). The broader serving matrix crosses full, KV-only, state-only, and joint precision with Random and ShareGPT workloads (3 seeds; 144 cells and 320,400 requests per attempt). Its frozen Gate 4 compares all 720 cell-threshold goodput values at a strict 10% run-stability tolerance.

5 RESULTS

5.1 Capacity

Table 1 and Figure 2 report the capacity measurements. Under int4 KV, bf16 state raises capacity by 37.6% (2B, 4K), 40.6% (9B, 4K), 11.5% (2B, 16K), and 14.0% (9B, 16K). The fp16-KV column shows smaller gains (7.0–15.4% at 4K) with mixed prediction signs. The discrete page-

Table 1. Capacity ratio r_{state} from bf16 state. Measured token capacity ratio and architecture-derived prediction; signed gap is (measured–predicted)/predicted. This 4K/16K slice illustrates allocator rounding; the full 52-pair matrix has residuals on both sides of zero, so the model is not a bound.

KV	Model	L	fp32 tok	bf16 tok	gap
int4	2B	4096	2,692,710	3,703,954	−2.37%
int4	2B	16384	4,895,837	5,458,458	−3.24%
int4	9B	4096	315,392	443,538	−0.18%
int4	9B	16384	573,440	653,635	−1.07%
fp16	2B	4096	1,199,383	1,384,448	+2.86%
fp16	2B	16384	1,552,143	1,661,337	+3.53%
fp16	9B	4096	144,104	161,899	−2.83%

block rounding is the source of the residual: fp32/bf16 block sizes change with layout (e.g., int4 2B: 2064/1072 tokens per block; fp16: 544/288). Across the complete 112-cell attempt, all 52 core pairs favor bf16 state; the median capacity gain is 15.44%. The model’s median absolute residual is 1.81%, with signed residuals from −3.66% to +13.21%. Because residuals cross zero, we use the model as an idealized predictor and make no bound claim. The clean second attempt reproduces all 112 cells and 52 directions within the frozen Gate 4 tolerances (maximum per-cell token difference 1.42%). Figure 1 shows the mechanism: bf16 state halves the per-block state footprint, which doubles the number of allocatable GPU blocks, and the measured ratio is the integer-arithmetic outcome of that re-partitioning.

Figure 2b also shows the compound KV ratio: with fp32 state, int4 KV yields 2.2451x over fp16 KV at 2B/4K; with bf16 state the same comparison yields 2.6754x. Both components draw from the same GPU pool but enter the per-sequence footprint as the additive terms AL and G ; changing either term changes the ratio of feasible capacities. In deployment terms, the fp32 GDN state at 2B is 18.63 MiB per sequence (9.32 MiB at bf16). At 4K under int4 KV, the measured total-token ceilings of 2,692,710 and 3,703,954 tokens correspond to 657 and 904 concurrent length-4096 sequences on the same GPU budget, so the state switch adds roughly 247 sequence slots without changing the KV bytes themselves.

5.2 Quality: perplexity and reasoning

Perplexity. Under int4 KV, the bf16-state marginal perplexity change is -0.0029 $[-0.0129, +0.0072]$ on C4 and $+0.0065$ $[-0.0447, +0.0578]$ on PG19 (3 seeds, paired 95% CI). Both intervals contain zero, and the magnitude is the same as the state change under fp16 KV ($-0.0003/+0.0004$), so stacking state-bf16 onto int4 KV introduces no measurable additional perplexity cost. The harness is a chunk-level approximation (Section 6); these numbers are supporting evidence for the kernel-level claim,

not a substitute for it. Figure 5 quantifies both statements: chunk-boundary rounding moves PPL by roughly 87% when the chunk size is reduced to one, while the marginal state cost under int4 KV stays within 0.007 of the fp16-KV cost on both corpora.

GSM8K. Figure 3 summarizes the 9-seed paired deltas. At 2B, bf16 state loses 1.00 point $[-1.71, -0.29]$, $p = 0.025$, with a pre-registered minimum detectable effect of 1.16 points and 67.5% observed power; this is a significant regression whose effect size is below the pre-registered MDE, and we report both. Across the nine paired dataset seeds the regression has paired Cohen’s $d = -0.92$; at 9B the paired $d = +0.54$, but the confidence interval contains zero, so we do not interpret it as a gain. Uniform int4 KV loses 2.72 points $[-4.20, -1.24]$, $p = 0.007$, power 88.3%. Stacking state-bf16 onto int4 KV shows a marginal +1.17 points $[-0.33, +2.66]$ relative to int4 alone: no additional regression. At 9B, bf16 state gains 0.33 points $[-0.07, +0.73]$, $p = 0.141$: no detectable regression. We do not aggregate the 2B and 9B effects. Figure 6 shows the same data at seed resolution: the nine thin lines pair each dataset seed across allocations, and the between-seed spread is what the confidence intervals in Figure 3 summarize.

RULER. Figure 4 reports the five cells (FWE and NIAH-multiquery, 4K/8K) that the earlier KV-quantization screen had singled out; the selection predates the state-dtype grid and is not a post hoc choice from these outcomes. Thinking is disabled to remove truncation of the model’s hidden reasoning from the 256-token answer budget. Across the three paired dataset seeds, fp32 and bf16 state produce identical accuracy in every cell: 91.11 and 98.33 for 2B FWE at 4K and 8K, and 100.00 for the three 9B cells. Thus all five bf16-minus-fp32 deltas are 0.00. An independent temporal attempt reproduces all 30 allocation-by-seed accuracy values and their sample-level predictions, references, hit vectors, and token counts exactly. The degenerate zero-width descriptive intervals reflect identical observed outputs, not population-level certainty; with only three paired seeds and 20 samples per cell, we do not claim equivalence or non-inferiority.

5.3 Per-layer sensitivity

We separately asked whether any single GDN layer should be kept at higher precision. Bf16 was applied to one GDN layer at a time (18 configurations, 2 corpora, 3 seeds, 36 tests). Two raw intervals excluded zero (C4, layers 2 and 8, $p = 0.049$ and $p = 0.0036$), but neither survives Bonferroni ($\alpha/36$) or BH-FDR correction. There is no evidence for a per-layer precision benefit, so the paper’s allocation dimension is the whole-state switch rather than layer-wise state precision (Figure 7).

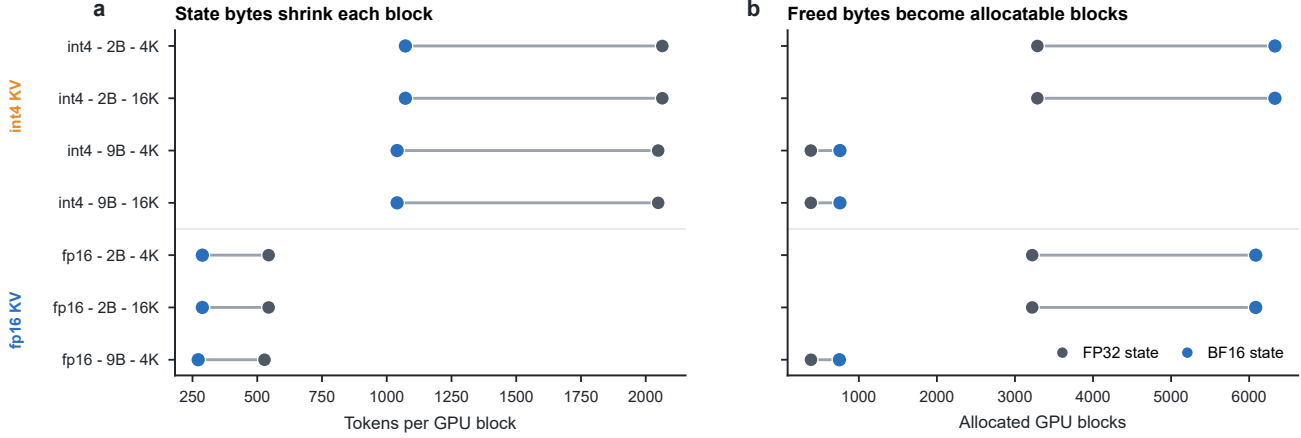


Figure 1. **Block granularity under fp32 vs. bf16 state.** (a) Tokens per GPU block and (b) number of allocated GPU blocks for the seven capacity cells. bf16 halves the per-block state footprint; the discrete re-partitioning of blocks explains the residual between measured and modeled r_{state} in Table 1.

5.4 Executable budget selection

The controller turns the paper’s two-dimensional accounting into an executable decision. It selects `full`, `state_only`, or `joint` for the strict, medium, and high frozen budgets, respectively; an infeasible request fails closed. Every selected mapping launches the corresponding vLLM precision arguments and meets the requested 500 ms TTFT and 200 ms TPOT SLO in all three confirmatory seeds. In the independent temporal run, mean goodput is 29.554, 29.552, and 38.905 req/s for strict, medium, and high; mean P95 TTFT is 194.23, 192.28, and 372.42 ms, and mean P95 TPOT is 18.04, 17.84, and 38.77 ms. All nine parent-to-rerun metrics fall within the predeclared environment-sensitive tolerances (10% for goodput and 20% for tail latency), with 18,000/18,000 requests and no failures in either attempt. This verifies policy execution for one Qwen3.5-2B, 4K, Random-workload, single-GPU slice. It is not evidence that the selected allocation dominates baselines across models or workloads.

5.5 Serving

Table 2 reports sustainable arrival rates for `int4` and `int4+bf16` allocations under the v3 protocol, and Figure 8 shows paired goodput deltas in the overload region.

The only repeated boundary difference is Random60 at 1000 ms (35 vs. 40 in the formal matrix), and it did not reappear in the second run (`int4` also reached 40 there). ShareGPT at 250 ms shows opposite directions between the two runs. We therefore do not present either workload as improved; the boundary tables are measurements, and we report them without assigning an effect.

What is stable across the two runs is the paired goodput delta

Table 2. **Sustainable rate boundaries (req/s, 3 seeds, protocol v3).** Formal matrix; values in parentheses are the second formal run (identical contracts and seeds). Differences between runs at Random60/1000 ms and ShareGPT/250 ms show that single boundary cells are run-sensitive and are not claimed as effects.

Workload	Config	250	500	1000	2000	3000
Random60	int4	30	35	35 (40)	40	40
Random60	int4+bf16	30	35	40	40	40
ShareGPT	int4	40 (35)	40	40	40	40
ShareGPT	int4+bf16	35 (40)	40	40	40	40

in the Random60 overload region; because the second run uses identical contracts and seeds, this establishes run-to-run stability rather than an independent sample. At r40, bf16 state improves goodput by 0.334 [0.078, 0.589] at 250 ms (paired t-test $p = 0.030$) and 0.215 [0.154, 0.276] at 500 ms ($p = 0.004$); the second formal run measures 0.304 [0.250, 0.359] ($p = 0.002$) and 0.138 [0.082, 0.195] ($p = 0.009$). At r45/r50 and 2000–3000 ms the deltas are smaller but consistently positive in both runs (e.g., r45/3000 ms: 0.367 [0.308, 0.426] vs. 0.372 [0.310, 0.435]; $p = 0.001$ and $p = 0.002$). None of the 60 serving cells survives BH-FDR at $q < 0.05$ (closest: formal r40/500 ms $q = 0.066$; second run r40/250 ms $q = 0.052$). We therefore treat the overload-region deltas as directional evidence whose weight comes from same-cell stability across the two runs, and we stop short of treating them as sustainable-SLO improvements. A cell-level direction audit over the same 60-cell grid is workload-limited: all 13 Random60 cells that are overloaded in both runs have positive deltas in both runs (13/13), while 7 of the 10 ShareGPT overload cells have opposite signs, so the directional evidence does not generalize across workloads. Per-cell p and BH q values for all 60

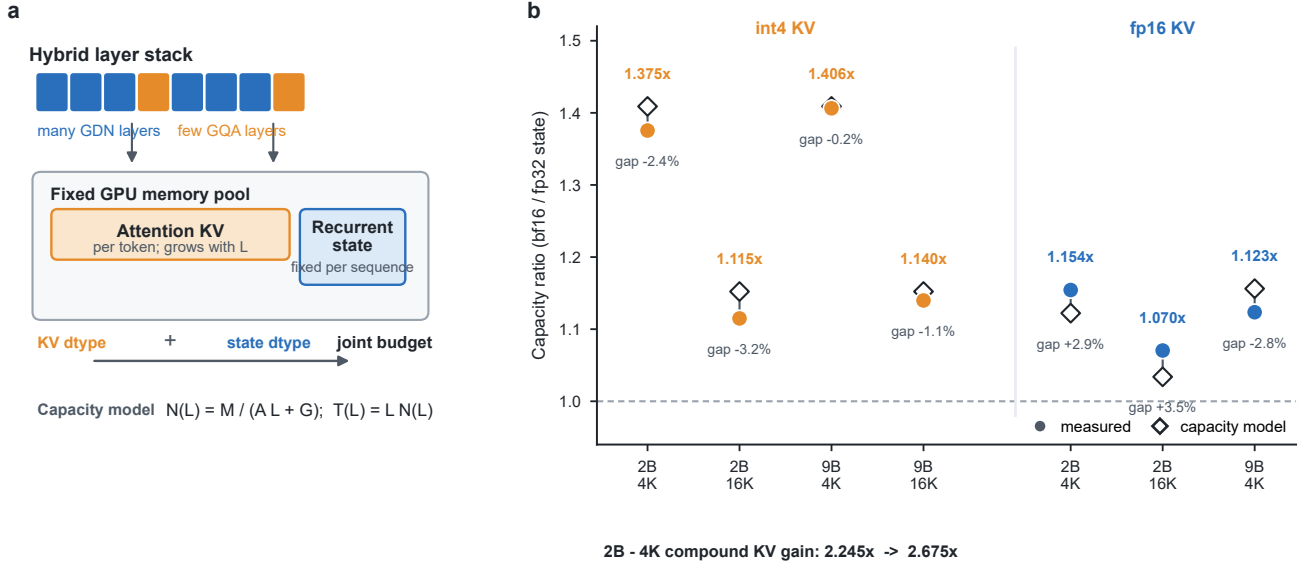


Figure 2. Precision budget for hybrid serving. (a) GDN recurrent state (fixed per sequence) and attention KV (growing with context) both draw from the same GPU pool; both bit-width (KV) and state precision are allocatable. (b) Measured vs. predicted r_{state} for the int4/fp16 KV columns (7 cells; gap labels are signed percent error). int4 column is a representative slice; the full matrix contains residuals of both signs.

cells are archived with the analysis artifacts. We report the goodput deltas as aggregate effects and do not assign them to a mechanism. The controlled contrasts below delimit which serving metrics reproduce. ShareGPT at 500 ms and above shows boundary parity in both runs.

The mechanism matrix completes all 18 allocation-by-seed cells and 5,400 requests in each attempt. Throughput reproduces within 10% in 18/18 cells (maximum relative difference 5.10%), but TTFT P95 does so in only 10/18 cells (maximum 75.39%) and TPOT P95 in 17/18 (maximum 12.13%). In the formal attempt, joint-minus-full mean throughput is -0.760 , -0.815 , and -2.492 req/s under fixed block count, fixed bytes, and fixed concurrency, respectively; none of the nine metric-by-contrast tests survives Holm correction. We therefore promote only the tested throughput direction and do not attribute a universal causal mechanism or exact latency effect.

Finally, the four-configuration formal matrix completes 144/144 cells and 320,400/320,400 measurement requests with zero failures in each of two runs. Nevertheless, its frozen Gate 4 fails: only 537/720 continuous goodput values are within 10% across attempts, 183/720 are outside tolerance, and the maximum symmetric relative difference is 79.61%. Sustainable labels are more stable (713/720 cell-threshold labels and 38/40 all-seed boundaries match), but these binary agreements do not override the failed primary gate. We therefore use this matrix to disclose serving instability, not to claim a four-configuration performance

advantage.

6 LIMITATIONS

Harness boundary. The perplexity harness rounds the state at chunk boundaries (chunk 128). A chunk-1 ablation changes PPL from 19.35 to 36.16 (+87%), which shows the harness is not equivalent to per-token kernel semantics. We treat PPL as supporting evidence and place the kernel-path quality conclusions (GSM8K, RULER) on the vLLM serving path.

RULER resolution. The repaired protocol disables thinking and has 20 samples in each of five cells with three paired dataset seeds. Exact agreement in both attempts is transparent but low-resolution evidence: it does not define an equivalence margin or establish equivalence, non-inferiority, or quality preservation outside the tested cells.

Generalization. All evidence is from Qwen3.5-2B/9B on one RTX 5090. Claims are scoped to GDN-based hybrids; Mamba2-style architectures and other hardware are untested. fp16-state quality is smoke-tested only; fp8/int8 state precisions are future work, so we make no claims about a full precision spectrum.

Serving variability. Overload-region goodput is stable in direction but not in magnitude across runs; boundary cells are run-sensitive. A workload with very long requests or a different mix may not show the effect. The broader four-

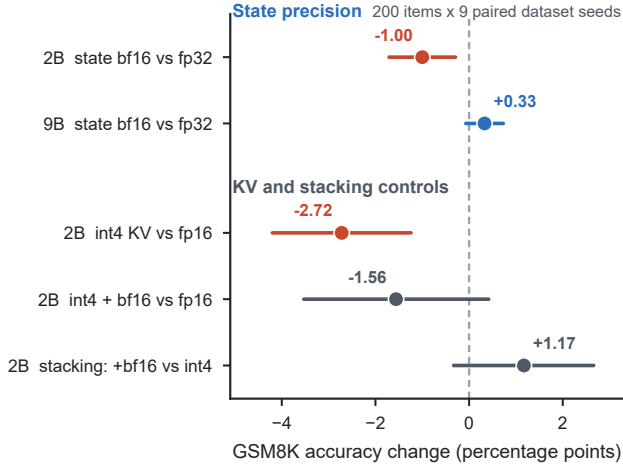


Figure 3. GSM8K paired deltas. 200 seeded items, greedy, no chain-of-thought, 9 dataset seeds; points and 95% paired CIs. Red intervals exclude zero. The 2B state and int4 regressions are significant; the 2B state effect is below the pre-registered MDE (1.16 pp) with 67.5% power.

configuration matrix does not pass its continuous-goodput stability gate. We did not test offloading or prefix caching, each of which changes the memory equation, and the four internal precision configurations do not substitute for those external system baselines.

Mechanism attribution. We ran fixed-block-count, fixed-bytes, and fixed-concurrency contrasts, but only throughput passes the temporal reproducibility threshold in every cell. TTFT and one TPOT cell do not. Moreover, the fixed-concurrency contrast changes the queueing constraint and is not a pure dtype intervention. These results narrow the plausible mechanism but do not identify a universal causal channel.

Controller scope. The selector is executable and reproducible, but its verified evidence covers one Qwen3.5-2B, 4K, Random-workload, TP=1 slice and three preregistered budget requests. It uses calibrated profiles and cold restarts rather than online adaptation, and it does not interpolate to unmeasured strata. Cross-model, cross-workload, cross-context, and switching overhead claims require separate evaluation.

Tensor parallelism. The single-GPU ratios extend to tensor parallelism only under uniform sharding. If the GDN state is sharded by its hidden dimension and the attention KV by its KV heads, both $A \cdot L$ and G scale as $1/P$, so $r_{state}(L)$ is unchanged to first order; for example, at $P = 2$ with the 2B/int4 KV and $L = 4096$ configuration, both terms halve and the ratio is unchanged to first order. Per-rank block rounding and communication overhead can shift the measured ratio, and we did not test TP=2/4; the transfer is

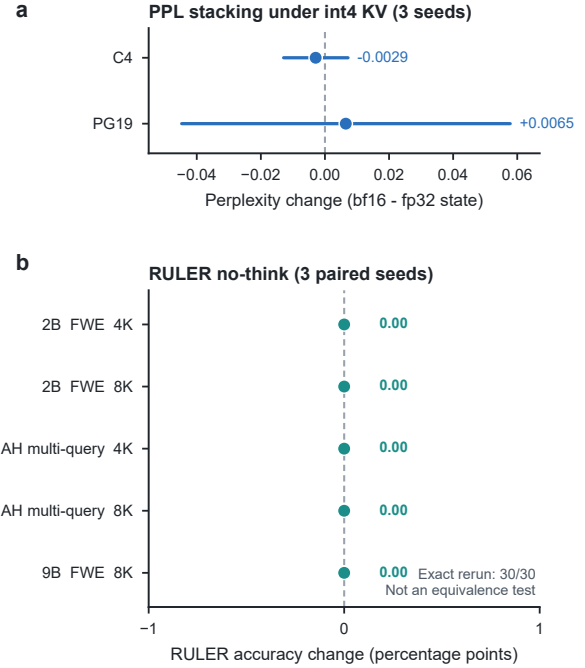


Figure 4. Perplexity stacking and RULER. (a) PPL marginal change of bf16 state under int4 KV (C4/PG19, 3 seeds). (b) RULER deltas (bf16 minus fp32 state) for the five screen-selected cells (FWE and NIAH-multiquery, 4K/8K); thinking disabled, 20 samples per cell, and 3 paired dataset seeds. Every observed delta is exactly zero in both attempts. Zero-width descriptive intervals are not an equivalence test.

an expectation, not a measurement.

Statistical resolution. GSM8K at 2B detects a 1.0-point regression with 67.5% power; effects smaller than ~ 1.16 points are below the pre-registered MDE. Serving cells use $n = 3$ seeds per cell and 60-cell BH-FDR control; no cell survives $q < 0.05$, so serving claims remain directional. The GSM8K seed resolution is $n = 9$ paired dataset seeds. The two serving runs share contracts and seeds, so the second run provides run-stability evidence rather than an independent sample. RULER uses only three paired dataset seeds; its exact zero deltas are descriptive and do not supply an equivalence bound. The M3 latency endpoints and M4 continuous-goodput family fail their frozen reproducibility gates and remain unverified.

7 CONCLUSION

Hybrid linear-attention serving has two allocatable precision dimensions: the attention-KV bit-width and the recurrent-state dtype. We measured the capacity, quality, and serving consequences of treating both as a joint budget and implemented a constraint-aware selector that produces executable

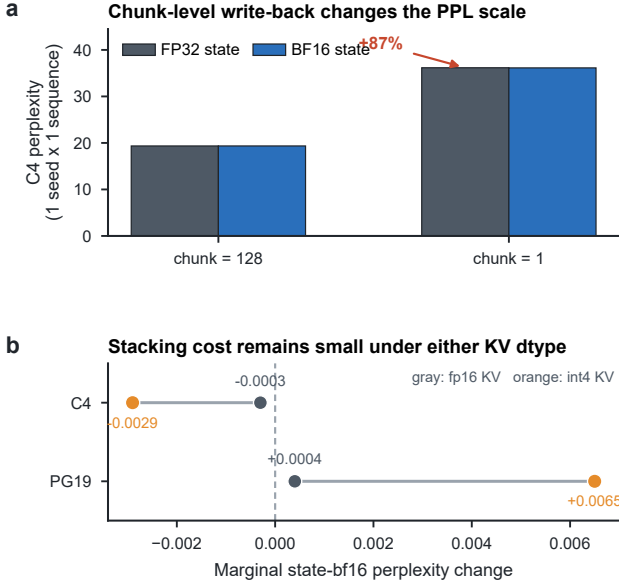


Figure 5. Harness boundary and stacking cost. (a) Chunk-level write-back rounding: chunk=1 raises C4 PPL by roughly 87% relative to chunk=128 (1 seed, 1 sequence), so PPL is a chunk-level approximation. (b) Marginal state-bf16 PPL delta under fp16 KV vs. int4 KV (3 seeds): stacking introduces no measurable additional cost.

deployments. The verified result is the capacity effect: all 52 core pairs in the 112-cell matrix favor bf16 state, with a 15.44% median gain. The capacity model is an idealized predictor with measured residuals, not a bound. Quality remains task-specific: the 2B GSM8K regression is retained, while no-think RULER shows exact observed agreement only in the five tested cells. The controller’s three budget mappings reproduce in one scoped slice. Controlled serving contrasts reproduce throughput but not tail latency, and the broader four-configuration matrix fails its continuous-goodput stability gate.

Operational rule. Under the target regime (short-context, high-concurrency, memory-bound serving), the state switch is material: at 2B/4K/int4 KV it adds roughly 247 concurrent sequence slots without changing the KV bytes themselves. When the KV footprint dominates the budget (e.g., at longer contexts), the relative gain shrinks, and operators should re-evaluate with the same capacity and quality protocol before adopting the switch.

We release the full measurement contracts, per-seed data, and reproduction artifacts so that the boundary cells and null results can be audited independently. Long-form and retrieval evidence beyond GSM8K and RULER (e.g., Long-Bench (Bai et al., 2024)) remains single-seed pilot evidence and is excluded from the claims above.

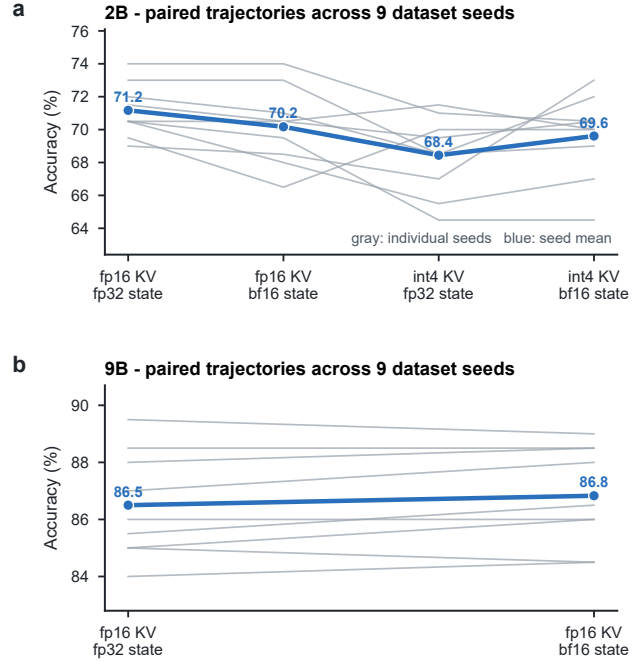


Figure 6. GSM8K per-seed paired accuracy. 200 seeded items, greedy, no chain-of-thought, 9 dataset seeds. Thin lines connect the same seed across allocations; the blue line and markers are seed means. (a) 2B across four allocations; (b) 9B across two. This figure is the per-seed resolution of Figure 3.

REFERENCES

- Bai, Y., Lv, X., Zhang, J., Lyu, H., Tang, J., Huang, Z., Du, Z., Liu, X., Zeng, A., Hou, L., Dong, Y., Tang, J., and Li, J. Longbench: A bilingual, multitask benchmark for long context understanding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 3119–3137, 2024.
- Botev, A., De, S., Smith, S. L., Fernando, A., Muraru, G.-C., Haroun, R., Berrada, L., Pascanu, R., Sessa, P. G., Dadashi, R., Hussenot, L., Ferret, J., Girgin, S., Bachem, O., Andreev, A., Kenealy, K., Mesnard, T., Hardin, C., Bhupatiraju, S., Pathak, S., Sifre, L., Rivière, M., Kale, M. S., Love, J., Tafti, P., Joulin, A., Fiedel, N., Senter, E., Chen, Y., Srinivasan, S., Desjardins, G., Budden, D., Doucet, A., Vikram, S., Paszke, A., Gale, T., Borgeaud, S., Chen, C., Brock, A., Paterson, A., Brennan, J., Risdal, M., Gundluru, R., Devanathan, N., Mooney, P., Chauhan, N., Culliton, P., Martins, L. G., Bandy, E., Huntsperger, D., Cameron, G., Zucker, A., Warkentin, T., Peran, L., Giang, M., Ghahramani, Z., Farabet, C., Kavukcuoglu, K., Hassabis, D., Hadsell, R., Teh, Y. W., and de Freitas, N. Recurrentgemma: Moving past transformers for efficient open language models. *arXiv preprint arXiv:2404.07839*, 2024.

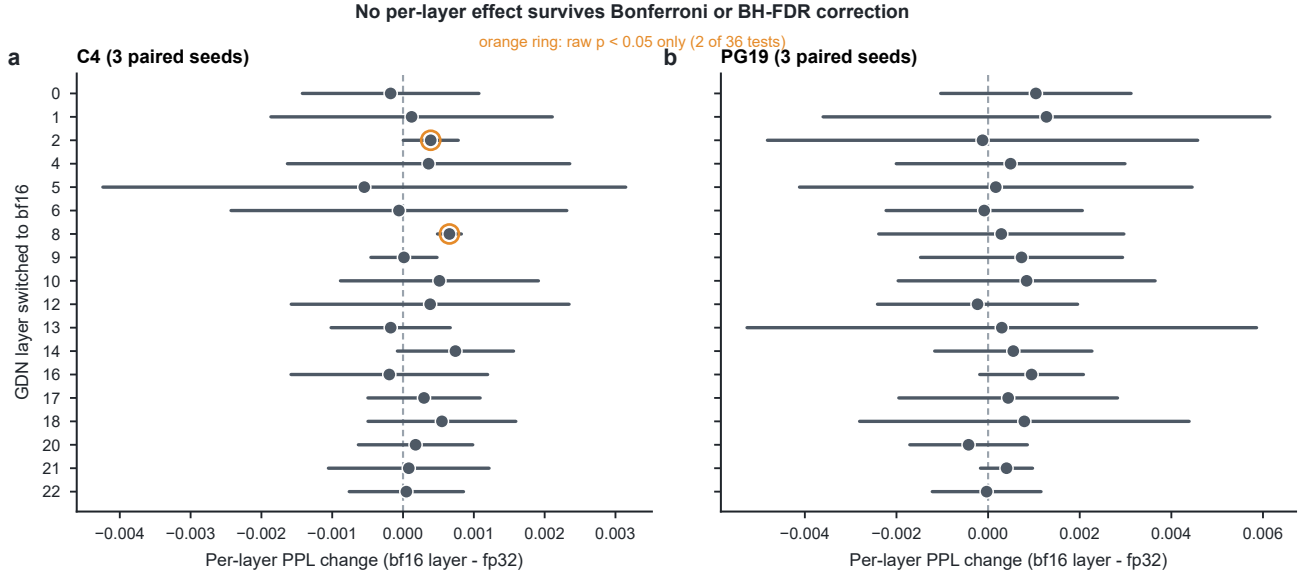


Figure 7. Per-layer state sensitivity (2B, C4/PG19, 3 seeds). Each row is bf16 applied to one GDN layer; points and paired 95% CIs are deltas against the fp32 baseline. Orange rings mark the two C4 layers with raw $p < 0.05$; neither is significant after Bonferroni or BH-FDR correction.

Chiang, H.-Y., Chang, C.-C., Frumkin, N., Wu, K.-C., and Marculescu, D. Quamba: A post-training quantization recipe for selective state space models. *arXiv preprint arXiv:2410.13229*, 2024.

Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., and Schulman, J. Training verifiers to solve math word problems. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

Dao, T. Replayssm: Cache ssm inputs, not state. *Technical blog*, 2026.

Dao, T. and Gu, A. Transformers are ssms: Generalized models and efficient algorithms through structured state space duality. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.

De, S., Smith, S. L., Fernando, A., Botev, A., Cristian-Muraru, G., Gu, A., Haroun, R., Berrada, L., Chen, Y., Srinivasan, S., Desjardins, G., Doucet, A., Budden, D., Teh, Y. W., Pascanu, R., De Freitas, N., and Gulcehre, C. Griffin: Mixing gated linear recurrences with local attention for efficient language models. *arXiv preprint arXiv:2402.19427*, 2024.

Glorioso, P., Anthony, Q., Tokpanov, Y., Whittington, J., Pilaault, J., Ibrahim, A., and Millidge, B. Zamba: A compact 7b ssm hybrid model. *arXiv preprint arXiv:2405.16712*, 2024.

Gu, A. and Dao, T. Mamba: Linear-time sequence modeling with selective state spaces. In *Conference on Language Modeling (COLM)*, 2024.

Hooper, C., Kim, S., Mohammadzadeh, H., Mahoney, M. W., Shao, Y. S., Keutzer, K., and Gholami, A. Kvquant: Towards 10 million context length llm inference with kv cache quantization. *arXiv preprint arXiv:2401.18079*, 2024.

Hsieh, C.-P., Sun, S., Krizan, S., Acharya, S., Rekesh, D., Jia, F., Zhang, Y., and Ginsburg, B. Ruler: What’s the real context size of your long-context language models? In *Conference on Language Modeling (COLM)*, 2024.

Kwon, W., Li, Z., Zhuang, S., Sheng, Y., Zheng, L., Yu, C. H., Gonzalez, J. E., Zhang, H., and Stoica, I. Efficient memory management for large language model serving with pagedattention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, 2023.

Lei, J. and Ilager, S. Arkv: Adaptive and resource-efficient kv cache management under limited memory budget for long-context inference in llms. In *Proceedings of the 26th IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGRID)*, 2026.

Li, Y., Huang, Y., Yang, B., Venkitesh, B., Locatelli, A., Ye, H., Cai, T., Lewis, P., and Chen, D. Snapkv: Llm knows what you are looking for before generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

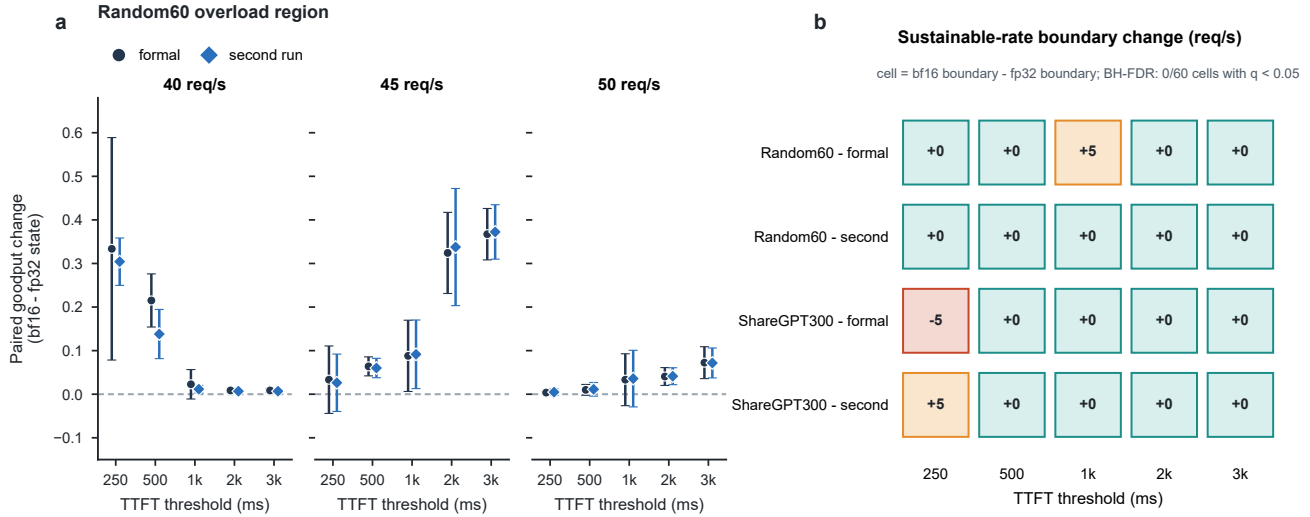


Figure 8. Serving, Random60 overload region. (a) Paired goodput delta (bf16 minus fp32 state) at rates 40–50 req/s, all thresholds, formal and second run, 3 seeds each, 95% CI. (b) bf16-minus-fp32 sustainable-rate boundary changes for both workloads and runs. The three nonzero cells are run-sensitive; none of the 60 cellwise comparisons survives BH-FDR.

Lieber, O., Lenz, B., Bata, H., Cohen, G., Osin, J., Dalmedigos, I., Safahi, E., Meirom, S., Belinkov, Y., Shalev-Shwartz, S., Abend, O., Alon, R., Asida, T., Bergman, A., Glozman, R., Gokhman, M., Manevich, A., Ratner, N., Rozen, N., Shwartz, E., Zusman, M., and Shoham, Y. Jamba: A hybrid transformer-mamba language model. *arXiv preprint arXiv:2403.19887*, 2024.

Liu, Z., Yuan, J., Jin, H., Zhong, S., Xu, Z., Braverman, V., Chen, B., and Hu, X. Kivi: A tuning-free asymmetric 2bit quantization for kv cache. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. URL <https://arxiv.org/abs/2402.02750>.

LMSYS. Sharegpt vicuna unfiltered dataset. Hugging Face dataset repository (lmsys/sharegpt_vicuna_unfiltered), 2023.

MLSys. Mlsys 2026 call for artifact evaluations. <https://mlsys.org/Conferences/2026/CallForAEs>, 2026.

Sharma, A., Ding, H., Li, J., Dani, N., and Zhang, M. Minikv: Pushing the limits of llm inference via 2-bit layer-discriminative kv cache. *arXiv preprint arXiv:2411.18077*, 2024.

Team, A. Q. Qwen3.5 model card (hybrid gated deltanet + gqa architecture). Hugging Face model repository, 2026.

vLLM Project. vllm pull request 22196: Mamba models – support fp32 ssm cache (mamba.ssm.cache_dtype). <https://github.com/vllm-project/vllm/pull/22196>, 2025.

vLLM Project. vllm pull request 43518: Fp8 ssm cache checkpointing (flashinfer ssu). <https://github.com/vllm-project/vllm/pull/43518>, 2026.

Wang, H., Gu, Y., Li, F., Wang, Z., Liu, Z., Wang, N., Li, X., and Yu, G. Hqekv: Towards hybrid quantization and eviction for kv cache in long-context llm inference. In *Findings of the Association for Computational Linguistics: ACL 2026*, 2026.

Xiao, G., Tian, Y., Chen, B., Han, S., and Lewis, M. Efficient streaming language models with attention sinks. In *International Conference on Learning Representations (ICLR)*, 2024.

Xu, Z., Yue, Y., Hu, X., Yuan, Z., Jiang, Z., Chen, Z., Yu, J., Xu, C., Zhou, S., and Yang, D. Mambaquant: Quantizing the mamba family with variance aligned rotation methods. *arXiv preprint arXiv:2501.13484*, 2025.

Yang, S., Wang, B., Zhang, Y., Shen, Y., and Kim, Y. Parallelizing linear transformers with the delta rule over sequence length. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Yang, S., Kautz, J., and Hatamizadeh, A. Gated delta networks: Improving mamba2 with delta rule. In *International Conference on Learning Representations (ICLR)*, 2025.

Ye, Z., Chen, L., Lai, R., Lin, W., Zhang, Y., Wang, S., Chen, T., Kasikci, B., Grover, V., Krishnamurthy, A., and Ceze, L. Flashinfer: Efficient and customizable

attention engine for llm inference serving. *arXiv preprint arXiv:2501.01005*, 2025.

Zandieh, A., Daliri, M., Hadian, M., and Mirrokni, V. Turboquant: Online vector quantization with near-optimal distortion rate. In *International Conference on Learning Representations (ICLR)*, 2026.

Zhang, J., Zhu, D., Song, Y., Wu, W., Kuang, C., Li, X., Shang, L., Liu, Q., and Li, S. More tokens, lower precision: Towards the optimal token-precision trade-off in kv cache compression. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025.

Zhang, J., Guo, H., Benini, L., and Li, Y. Rdkv: Rate-distortion bit allocation for joint eviction and quantization of the kv cache. *arXiv preprint arXiv:2605.08317*, 2026.

Zhang, Z., Sheng, Y., Zhou, T., Chen, T., Zheng, L., Cai, R., Song, Z., Tian, Y., Ré, C., Barrett, C. W., Wang, Z., and Chen, B. H2o: Heavy-hitter oracle for efficient generative inference of large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

A CAPACITY MODEL DERIVATION

For Qwen3.5-2B, the GDN state per sequence is derived from the vLLM `MambaStateShapeCalculator` path: 18 layers $\times$ 1,085,440 bytes per layer at fp32 (code-derived; the 9B state is 24.84 MiB). The per-token attention-KV footprints A_f and A_q are computed from the GQA head/layer configuration and the vLLM packed page layout (fp16: 2 bytes per element; int4: 0.5 bytes per element). G for bf16 is half of G_{fp32} . Eq. (2) uses these derived values; the predicted ratios reported in Table 1 are evaluated with no fitted parameters. The observed residual is attributable to the allocator’s discrete block rounding, which we verified from the per-attempt block-size and block-count records.

For Eq. (1), $N(L)$ is the number of concurrent length- L sequences and $T(L)$ is their total token capacity.